

# Capstone Project — Full-Stack Web App

SESSION 1: BECOMING A FULL-STACK CREATOR





# Be the CEO of Your Own App Startup

Imagine creating an app that connects neighbors to share **delicious home-cooked meals**. How would you design and build this innovative solution?

# Understanding Full-Stack Development: Frontend and Backend

## THE WHOLE PACKAGE

Full-Stack development combines the **visible Frontend** that users interact with and the **hidden Backend** that manages data and logic effectively.





## HTML/CSS

The **building blocks** of the Frontend design.



## WEBSOCKETS

Provide **real-time updates** like live delivery tracking.



## CLOUD STORAGE

**Saving your data** safely online for easy access.



## DATABASE

The digital '**Godown**' storing all your important info.



## API

The 'Waiter at a roadside restaurant' carrying messages between Frontend and Backend.

# Technologies Working Together

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# Recap: Understanding the MVC Pattern

## THE DIGITAL BLUEPRINT

The MVC pattern consists of three key components: **View**, which is the user interface; **Model**, which manages data; and **Controller**, which handles logic.



# Community App Ideas

## 01 SOCIETY HELPER

Track gate entries and deliveries for easier management.

## 02 STUDY BUDDY

Create a chat platform for real-time homework assistance.

## 03 LASSI STALL TRACKER

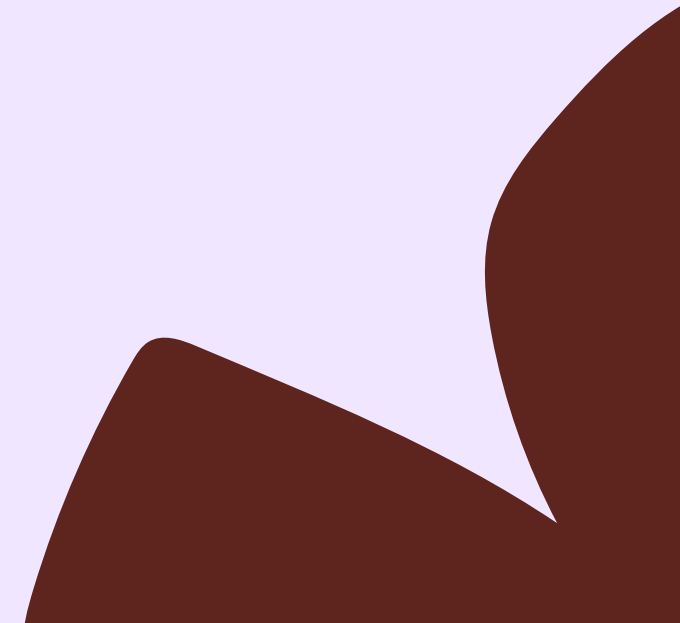
Locate nearby lassi stalls and their special offers.

## 04 ECO-WARRIOR

Log plastic recycling efforts within your community efficiently.

## 05 RWA NOTICE BOARD

Share important announcements and updates for residents easily.



# Real Indian Community Apps



## SOCIETY HELPER

Helps track gate entries or water delivery.



## ECO-WARRIOR

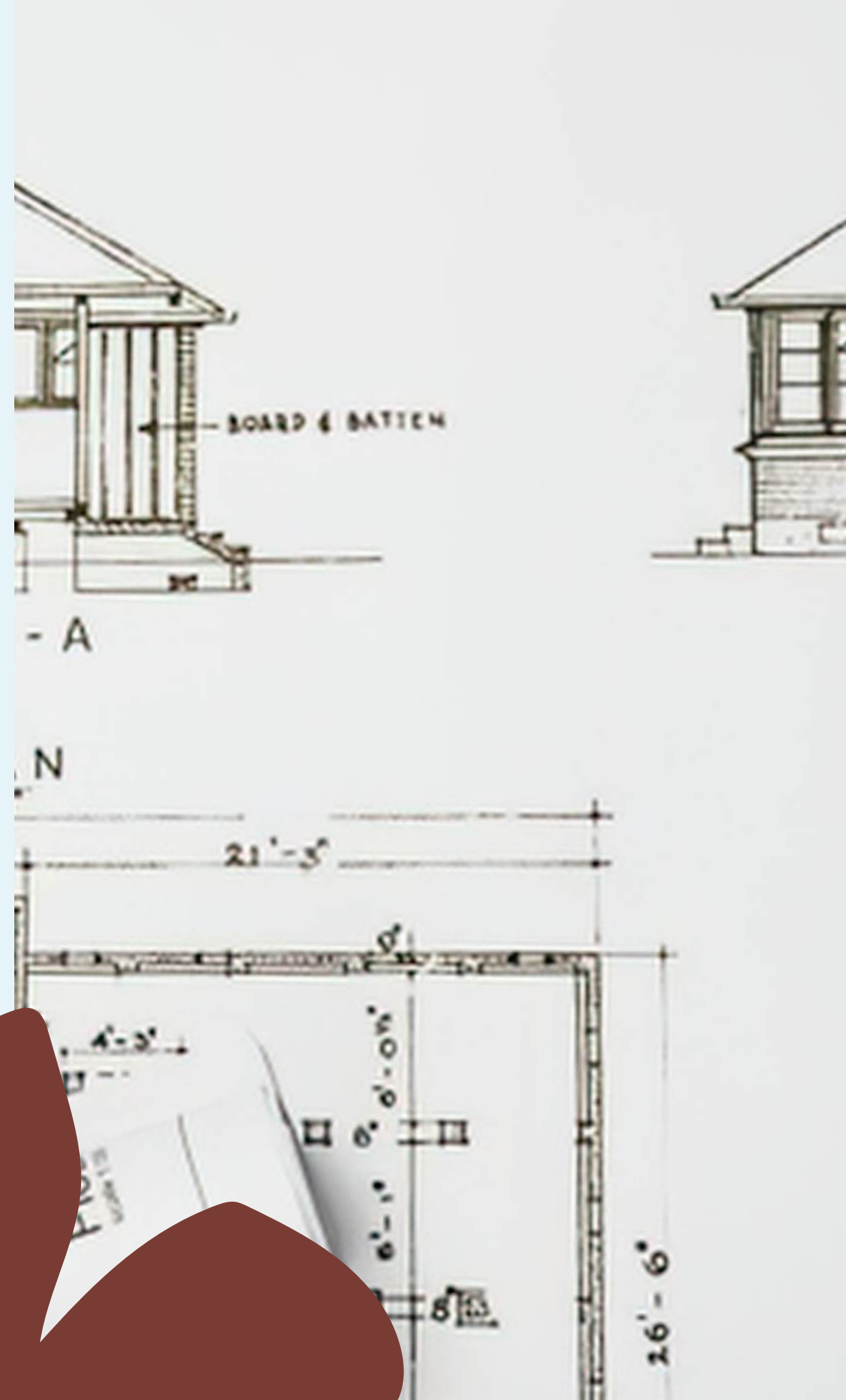
Logs plastic recycling efforts in the colony.



## STUDY BUDDY

Provides real-time chat for homework assistance.

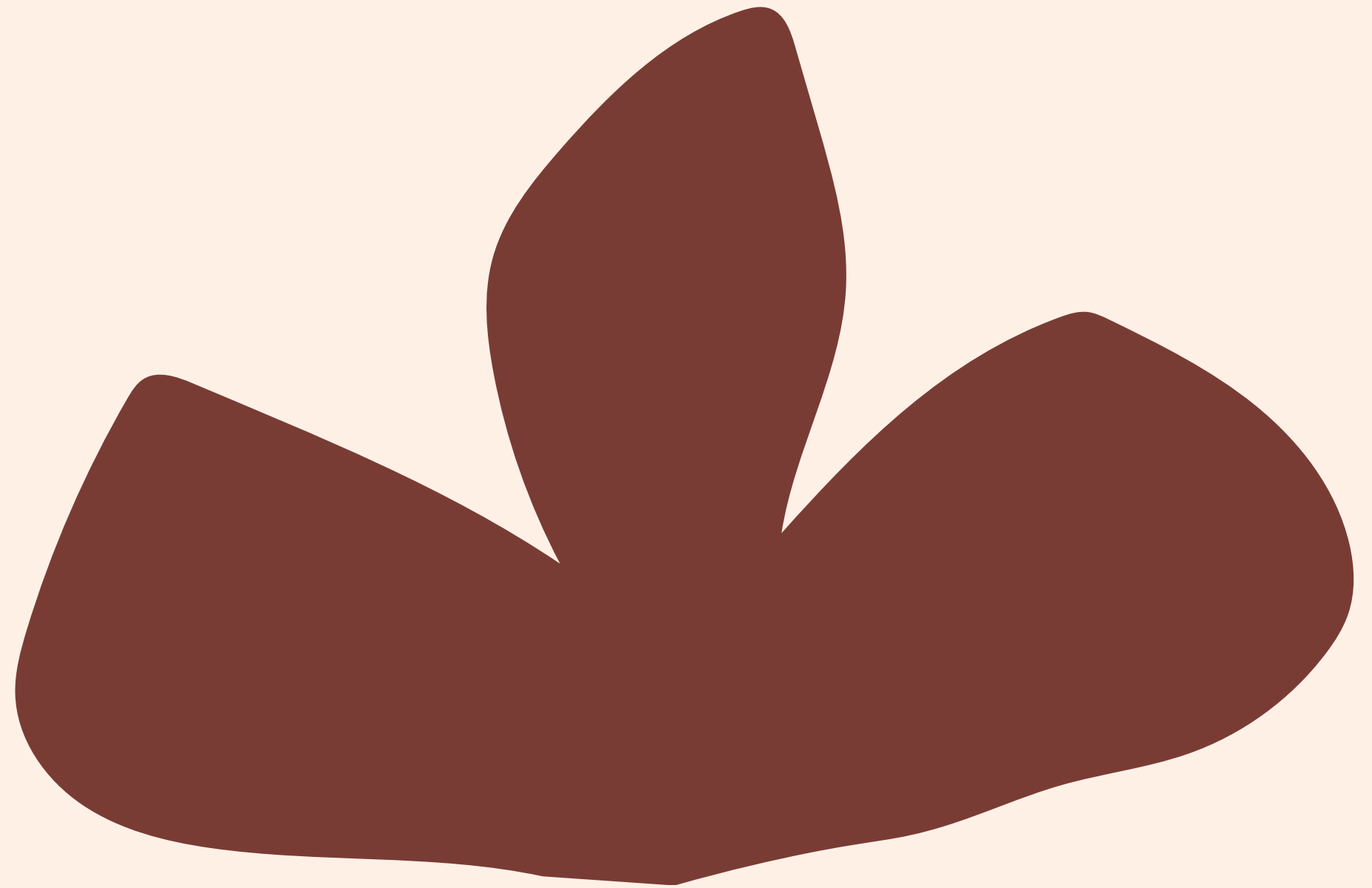
# Your App's Digital Blueprint



# Draw Your App Blueprint

## PLAN YOUR APP'S ARCHITECTURE HERE

1. What will users see? (Frontend)
2. What data will be saved? (Database)
3. Will it need live updates? (WebSockets)
4. Will you use Cloud storage?



# Indian Tech Success Stories



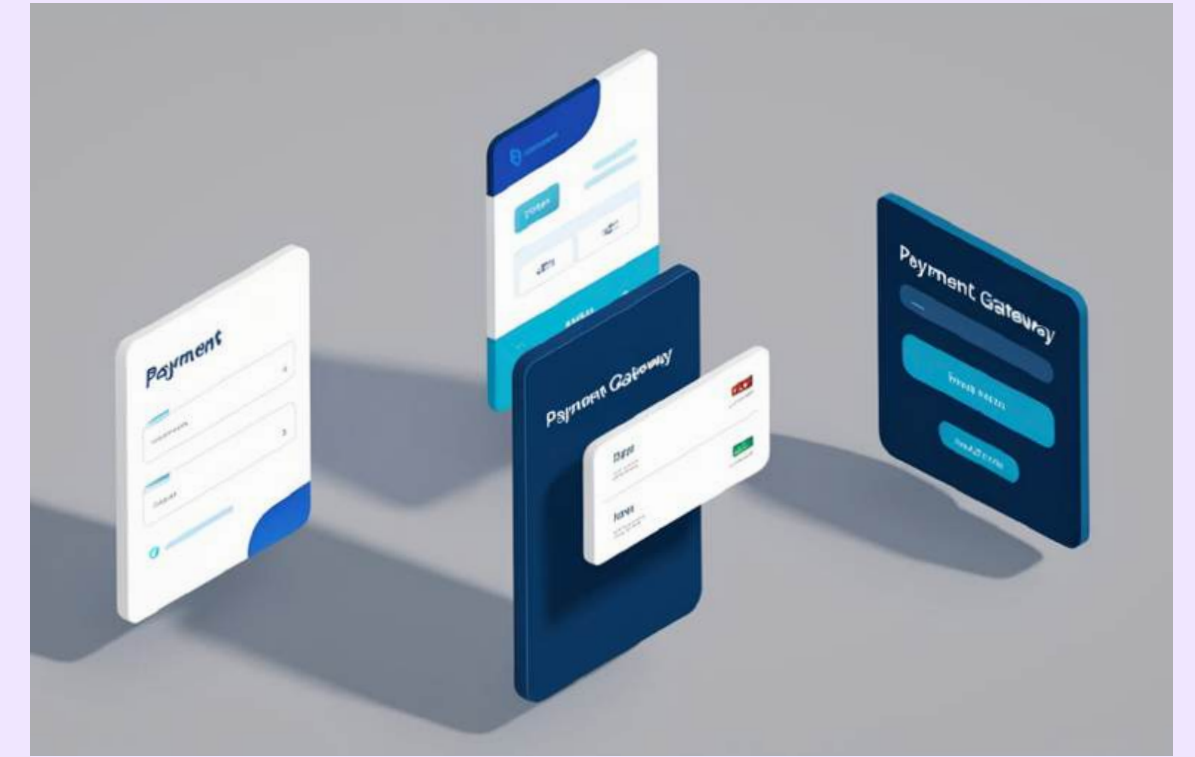
## ZOMATO

**Revolutionized** food delivery in India with great convenience.



## SWIGGY

Offers fast delivery from local restaurants to your door.



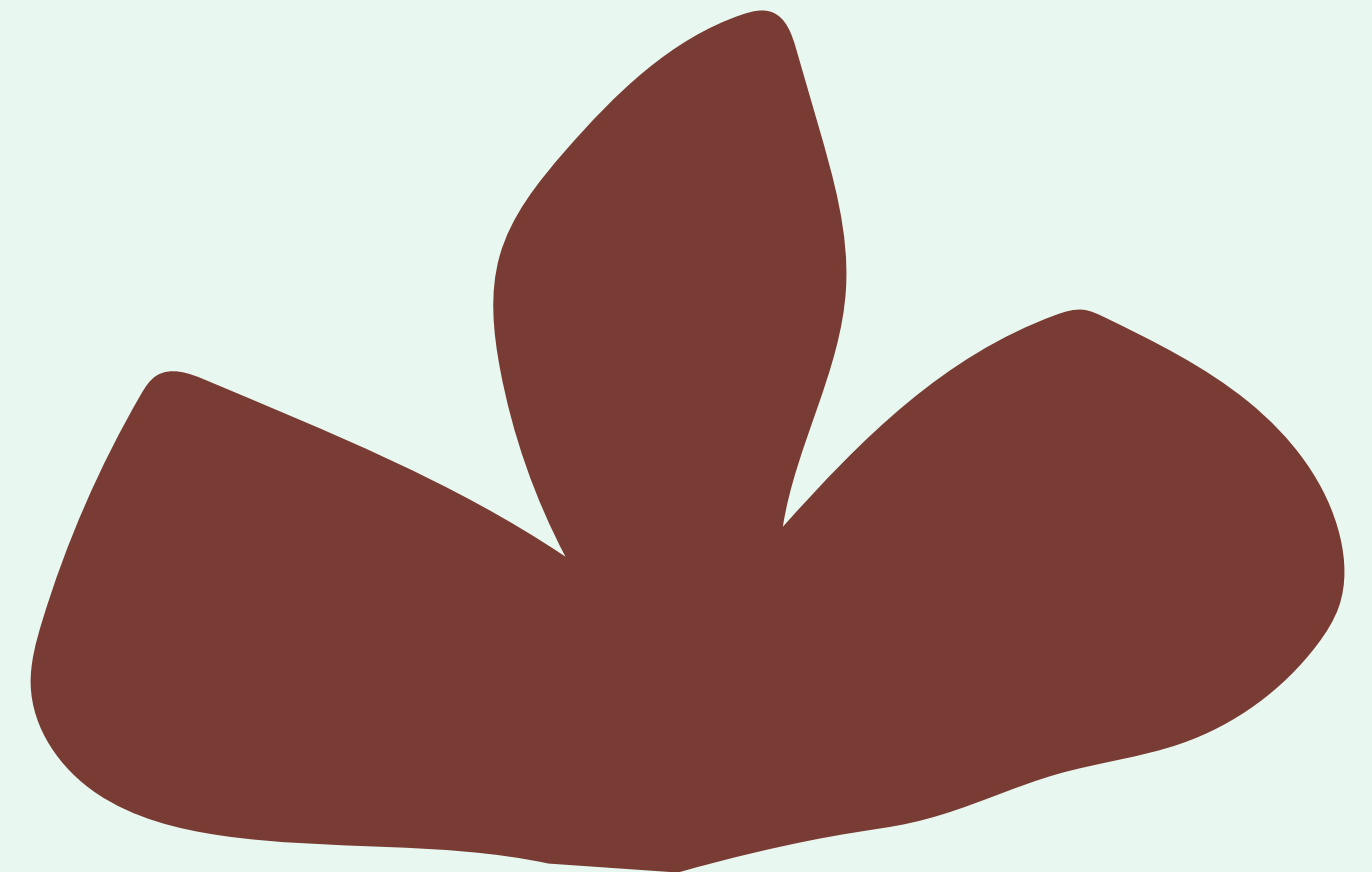
## PAYTM

Simplifies **digital payments** and online transactions for everyone.

# Key Takeaways from Session

## **UNDERSTANDING FULL-STACK DEVELOPMENT BASICS**

- Full-Stack combines Frontend and Backend
- Technologies like Cloud and MVC interact
- Apps can solve real community issues
- Planning your blueprint is essential



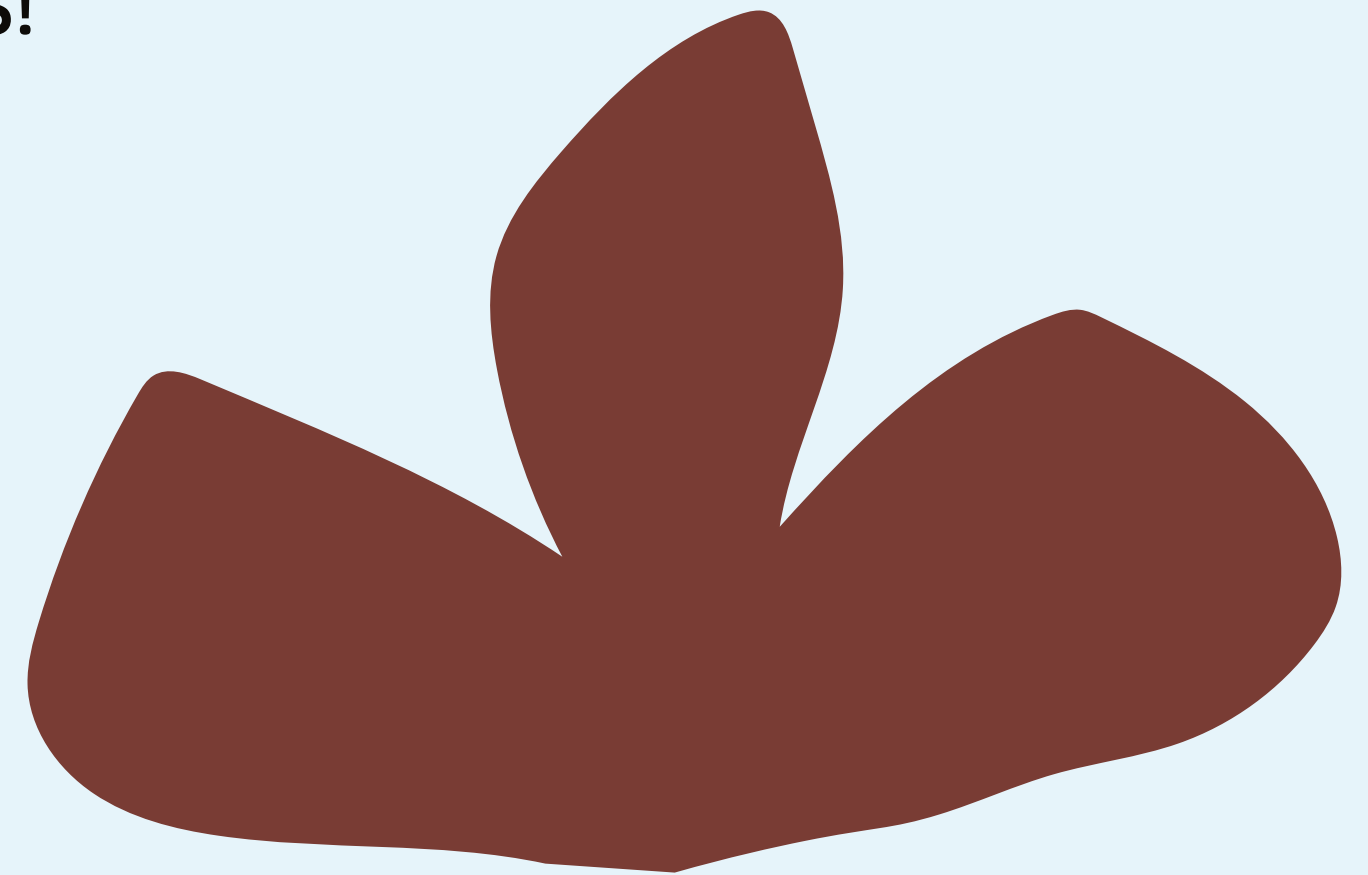


# Your Journey as a Junior Software Architect

You're on your way to becoming a **full-stack creator**! With each session, you'll build skills to turn ideas into real apps.

# Exciting App Ideas

SHARE YOUR THOUGHTS!





Thank You for Joining Us!